

ORC · LONG-RANGE CASIMIR SERIES · PAPER I

Electrostatic Patch Potentials in Long-Range Casimir Force Measurements

A Quantitative Noise Budget and Likelihood Framework

*Corrected pressures · $V^2\lambda^2$ degeneracy · Material-swap discriminator ($\Delta F/F < 2\%$ Au-vs-Cu)
Identifies the $d \sim 3\text{--}10\ \mu\text{m}$ sphere-plane danger zone for future long-range claims.*

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Paper I of three. Patches first; gravity gradients (Paper II) and thermal-mechanical (Paper III) follow.

On the question: in the regime where Casimir signals become extraordinarily weak, what backgrounds dominate?

ABSTRACT

We present a quantitative noise budget for Casimir force measurements at separations from $0.1\ \mu\text{m}$ to $10\ \text{mm}$, focusing on electrostatic patch potentials within an exponential surface-correlation model. The ideal-conductor zero-temperature pressure is $P_0 = \pi^2\hbar c/(240d^4)$ and the high-temperature thermal contribution is $P_T \approx \zeta(3)k_B T/(4\pi d^3)$; the latter dominates beyond $\lambda_T = \hbar c/(k_B T) \approx 7.6\ \mu\text{m}$ at $300\ \text{K}$. Within standard exponential patch correlation, patch contributions reach the order of the Casimir force in the $d \sim 3\text{--}10\ \mu\text{m}$ sphere-plane regime for typical room-temperature surface preparation ($V_{\text{rms}} \sim 5\ \text{mV}$, $\lambda_c \sim 1\ \mu\text{m}$). Outside this window — at sub-micron separations, in cryogenic large-radius geometries, or with highly-prepared surfaces — Casimir typically dominates. We re-examine published sub-micron and cryogenic experiments (Decca et al. 2005/2007, sphere-plane $R = 150\ \mu\text{m}$, $d = 750\ \text{nm}$; Sushkov et al. 2011, large-lens $R = 156\ \text{mm}$, $T = 4\ \text{K}$, $d = 0.7\text{--}7\ \mu\text{m}$) and find their published regimes are *not* patch-dominated under standard parameter assumptions; the analysis therefore does not contest those results, but identifies the parameter regime where future long-range claims will be vulnerable. We further demonstrate a likelihood degeneracy: force-data analyses constrain primarily the combination $V_{\text{rms}}^2\lambda_c^2$, not the individual parameters, requiring independent surface-potential characterization (KPFM) for unambiguous interpretation. **We propose a falsifiable material-swap test:** under standard Lifshitz theory the Casimir force should change by $< 2\%$ between Au and Cu at $d = 10\ \mu\text{m}$, while patch-dominated forces typically change by $20\text{--}200\%$ due to material-dependent surface-preparation histories. To our knowledge no published long-range Casimir experiment has reported this test; we recommend it as a prerequisite for future claims at $d > 3\ \mu\text{m}$.

1. Introduction

The Casimir force between uncharged plates [1,2] has been verified at sub-micron separations with percent-level agreement [3-5]. Beyond $\sim 3 \mu\text{m}$ two things change: the force becomes extraordinarily small ($P_0 \propto d^{-4}$), and the thermal-photon contribution dominates ($P_T \propto d^{-3}$). Real metal surfaces exhibit random work-function variations from grain boundaries, adsorbates, and oxides, producing electrostatic patch potentials with $\Delta V_{\text{rms}} = 1\text{-}10 \text{ mV}$ and lateral correlation lengths $\lambda_c = 0.1\text{-}5 \mu\text{m}$ in standard evaporated films [6-8].

This paper builds a complete electrostatic noise budget for long-range Casimir measurements and applies it to two representative published claims. We identify the parameter window in which patches dominate the Casimir signal, demonstrate a likelihood degeneracy that prevents unique parameter extraction from force data alone, and propose a falsifiable material-swap test that discriminates Casimir-dominated from patch-dominated regimes. This is Paper I of a three-paper series; Papers II (gravitational gradients and seismic noise) and III (thermal-mechanical mode coupling) treat the other two dominant noise domains.

2. Casimir Pressures and Forces at Long Range

2.1 Ideal-Conductor Expressions

For ideal parallel plates the pressures are

$$P_0(d) = \pi^2 \hbar c / (240 d^4), \quad P_T(d) \approx \zeta(3) k_B T / (4\pi d^3) \text{ for } d \gg \lambda_T$$

where $\lambda_T = \hbar c / (k_B T) \approx 7.6 \mu\text{m}$ at 300 K. Real metals (Au, Cu) deviate from the ideal-conductor limit at $d \lesssim \lambda_p \sim 100\text{-}150 \text{ nm}$; at $d \gg \lambda_p$ the perfect-conductor values apply to within percent-level corrections [4]. The plasma-Drude controversy modifies the thermal contribution by a factor of order unity [12], subdominant to the patch-induced variations discussed below.

2.2 Numerical Values for Parallel Plates (1 cm² area)

Table 1 lists numerically verified pressures and forces at four representative separations. All values were computed and cross-checked from the closed-form expressions above.

Separation	P ₀ (N/m ²)	P _T (N/m ²)	F ₀ (N) [1 cm ²]	F _T (N) [1 cm ²]	P _T /P ₀
1 μm	1.30×10^{-3}	3.96×10^{-4}	1.30×10^{-7}	3.96×10^{-8}	0.30
10 μm	1.30×10^{-7}	3.96×10^{-7}	1.30×10^{-11}	3.96×10^{-11}	3.0
100 μm	1.30×10^{-11}	3.96×10^{-10}	1.30×10^{-15}	3.96×10^{-14}	30
1 mm	1.30×10^{-15}	3.96×10^{-13}	1.30×10^{-19}	3.96×10^{-17}	305

3. Electrostatic Patch Potentials

3.1 Exponential-Correlation Model

For two surfaces with random potential $\phi(\mathbf{r})$ and exponential autocorrelation $\langle V(\mathbf{r})V(\mathbf{0}) \rangle = V_{\text{rms}}^2 \exp(-r/\lambda_c)$, the patch pressure has two asymptotic regimes [10,11]:

$$\begin{aligned} \text{Near field } (d \ll \lambda_c): P_{\text{patch}} &\approx \epsilon_0 V_{\text{eff}}^2 / (2 d^2) \\ \text{Far field } (d \gg \lambda_c): P_{\text{patch}} &\approx \epsilon_0 V_{\text{eff}}^2 \lambda_c^2 / (2 d^4) \end{aligned}$$

with $V_{\text{eff}}^2 = 2 V_{\text{rms}}^2$ for two statistically independent surfaces. The crossover between the two regimes is smooth and occurs over $0.5 \lambda_c \lesssim d \lesssim 5 \lambda_c$.

Spectral assumption: the exponential correlation captures short-range disorder from grain boundaries; macroscopic patches ($\lambda_c \sim 10\text{-}100 \mu\text{m}$ from temperature gradients, mounting

stress, adsorbed-charge inhomogeneities) extend the near-field regime to larger d and are discussed in §3.3. Alternative spectra (Gaussian, power-law) preserve the asymptotic scaling laws and the likelihood degeneracy structure of §6.

3.2 Patch-to-Casimir Ratios (Parallel Plates)

Table 2 reports the patch/Casimir-total pressure ratios numerically. Values use the smooth interpolation between near- and far-field expressions and include the thermal Casimir contribution.

Surface quality ($V_{\text{rms}}, \lambda_c$)	1 μm	10 μm	100 μm	1 mm
Best (1 mV, 0.2 μm)	0.001	$< 10^{-4}$	$< 10^{-5}$	$< 10^{-6}$
Good (3 mV, 0.5 μm)	0.024	0.004	5×10^{-4}	5×10^{-5}
Standard (5 mV, 1 μm)	0.13	0.04	0.005	6×10^{-4}
Poor (10 mV, 2 μm)	0.52	0.67	0.087	0.009

Headline finding: for parallel-plate geometry with standard surface preparation ($V_{\text{rms}} \sim 5$ mV, $\lambda_c \sim 1$ μm), patches contribute at the few-percent level throughout the long-range regime — non-negligible but not dominant. With *poor* surface preparation (10 mV, 2 μm), patches reach $\sim 67\%$ of the Casimir signal at $d \sim 10$ μm , in the crossover regime where neither near-field nor far-field asymptote applies cleanly.

3.3 Sphere-Plane Geometry and Macroscopic Patches

Most Casimir experiments use sphere-plane geometry (proximity force approximation, PFA: $F = 2\pi R P$) rather than parallel plates, since alignment is far easier. Under PFA the *ratio* of patch force to Casimir force is the same as for parallel plates, since both forces acquire identical $2\pi R$ prefactors. However, sphere-plane geometry is sensitive to additional near-field logarithmic enhancements not captured in the simple PFA expression [10,11], which can increase the patch force by $O(1)$ factors for $d < \lambda_c$.

Macroscopic patches ($\lambda_c \sim 10$ – 100 μm) from temperature gradients, mounting stress, or adsorbed-charge inhomogeneities can extend the near-field regime to substantially larger d . If $V_{\text{rms}} \sim 5$ mV with $\lambda_c \sim 50$ μm is present, the patch force at $d = 10$ μm can exceed the Casimir force by $O(10)$, and at $d = 100$ μm by $O(100)$. Such macroscopic patches are difficult to eliminate with standard surface preparation and may explain anomalously large backgrounds in some long-range experiments.

4. Why Single-Voltage Subtraction Fails

Most experiments fit the total force versus an applied bias V as

$$F_{\text{tot}}(V) = F_C + F_{\text{patch}} + \frac{1}{2} (dC/dz)(V - V_0)^2$$

and extract F_C at the parabola minimum. This procedure assumes a uniform contact-potential difference V_0 . For a spatially-varying patch landscape, the residual after subtracting the mean potential is the *variance* of the patch field, not zero. For a Gaussian random patch field with exponential correlation, the residual is approximately 60–90% of the original patch force, consistent with numerical simulations. The exact fraction depends on the spectral weight: surfaces with more power at small wavenumber (large-scale patches) retain higher residuals. No single V_0 can null a spatially varying potential.

5. Re-examination of Representative Long-Range Claims

We apply the noise budget to two of the most-cited long-range / sub-micron Casimir measurements to establish that the analysis correctly identifies which experiments are vulnerable to patches. **The conclusion is that neither published experiment was operating in a patch-dominated regime;** this strengthens those results rather than weakening them, and clarifies where future experiments venturing into $d \sim 3\text{--}10\ \mu\text{m}$ sphere-plane territory must take additional care.

5.1 Decca et al. (2005/2007): Sub-micron Sphere-Plane

Decca et al. [13,14] used a MEMS torsion oscillator with $R = 150\ \mu\text{m}$ sphere on a planar Au surface, at $d = 160\text{--}750\ \text{nm}$ and room temperature. Their reported $V_{\text{rms}} \approx 3\ \text{mV}$ and $\lambda_c \approx 0.5\ \mu\text{m}$ place the $d = 750\ \text{nm}$ endpoint at $d/\lambda_c = 1.5$, in the crossover regime. The patch/Casimir force ratio at this point is approximately 0.02, with order- $O(0.1)$ uncertainty arising from the crossover-regime ambiguity. Their single-voltage subtraction procedure removes the mean component, leaving a sub-percent residual variance — comfortably below the few-percent statistical and systematic errors quoted.

Without access to spatially-resolved KPFM data and the experiment's complete subtraction analysis, we cannot definitively re-derive their result. What this analysis shows is that under standard parameter assumptions, patch contributions of the order required to threaten their result are not supported. The Decca measurements remain the cleanest published sub-micron Casimir verification.

5.2 Sushkov et al. (2011): Cryogenic Large-Lens Geometry

Sushkov et al. [5] used a cryogenic torsion balance with a large gold-coated lens ($R \approx 156\ \text{mm}$) on a planar gold surface, at $d = 0.7\text{--}7\ \mu\text{m}$ and $T = 4\ \text{K}$. Cryogenic operation suppresses both the thermal Casimir contribution ($\lambda_T \approx 572\ \mu\text{m}$ at 4 K, so $P_T \ll P_0$ across the experiment's range) and the patch-potential drift (typically $\sim 0.5\text{--}2\ \text{mV/hour}$ reduces by $O(10)$ at cryogenic temperatures). At $d = 2\ \mu\text{m}$ with $V_{\text{rms}} \approx 2\ \text{mV}$ and $\lambda_c = 0.5\text{--}2\ \mu\text{m}$, the patch/Casimir force ratio computed via PFA is in the range 0.03–0.7, depending on λ_c .

For the conservative end of the parameter range, patches reach a few percent of the Casimir signal; for unfavorable $\lambda_c \sim 2\ \mu\text{m}$, patches can approach ~70% of the Casimir signal at $d = 2\ \mu\text{m}$. Sushkov reported 5% agreement with Lifshitz theory, which is consistent with the favorable end of the parameter range but sensitive to the assumed patch spectrum. *The experiment's spatially-resolved patch landscape was not characterized;* the result is robust if $\lambda_c \leq 1\ \mu\text{m}$ but more uncertain if larger correlation lengths are present. As with Decca, this analysis does not contest Sushkov's published result, but identifies the surface-characterization gap that future experiments should close.

5.3 Where Patches Will Dominate: The $d \sim 3\text{--}10\ \mu\text{m}$ Sphere-Plane Window

The danger zone for future long-range Casimir experiments is room-temperature sphere-plane operation at $d \sim 3\text{--}10\ \mu\text{m}$ with standard-grade surfaces. In this window, $V_{\text{rms}} = 5\ \text{mV}$, $\lambda_c = 1\ \mu\text{m}$ produces patch/Casimir ratios of 0.04–0.13 at the small end and is sensitive to crossover-regime uncertainty at the large end. With $V_{\text{rms}} = 10\ \text{mV}$, $\lambda_c = 2\ \mu\text{m}$, ratios reach 60–70% at $d = 10\ \mu\text{m}$. **Any future claim of long-range Casimir measurement in this window without spatially-resolved KPFM characterization should be treated as provisional.**

6. Likelihood Framework and the $V^2\lambda^2$ Degeneracy

Given an observed force $F_{\text{obs}}(d)$ with Gaussian errors σ_i , we define the log-likelihood under the Casimir + patch model as

$$\ln L(\boldsymbol{\theta}) = -\frac{1}{2} \sum_i [(F_{obs}(d_i) - F_C(d_i) - F_{patch}(d_i; \boldsymbol{\theta})) / \sigma_i]^2,$$

where $\boldsymbol{\theta} = (V_{rms}, \lambda_c)$. In the far-field regime $d \gg \lambda_c$, $F_{patch} \propto V_{rms}^2 \lambda_c^2 / d^4$: only the combination $V_{rms}^2 \lambda_c^2$ is constrained by force data, not the individual parameters. The likelihood contours in the (V_{rms}, λ_c) plane elongate along the curve $V_{rms}^2 \lambda_c^2 = \text{const}$, preventing unique parameter extraction.

Implication: patch-inclusive models can fit force data with widely varying (V_{rms}, λ_c) combinations, none uniquely identified. Independent characterization (KPFM scanning of the actual surfaces used) is therefore essential for unambiguous interpretation. Force data alone cannot exclude patch-dominated explanations once both parameters are allowed to vary within their physically plausible ranges.

7. A Falsifiable Material-Swap Test

7.1 Principle

At $d \gg \lambda_p \sim 145$ nm (Au plasma wavelength), real-metal corrections to the ideal-conductor Casimir force scale as $(\lambda_p/d)^2$. At $d = 10$ μm this is approximately 0.02% — far below the percent-level differences arising from Lifshitz integration over realistic optical data. The Casimir force difference between Au ($\hbar\omega_p = 8.55$ eV) and Cu ($\hbar\omega_p = 8.76$ eV) at $d = 10$ μm is therefore predicted to be at most 2%.

Patches behave entirely differently. Different metals have different grain structures, different work functions (affecting adsorbate binding), and different surface oxidation chemistries. The patch variance V_{rms} typically differs by factors of 2–10 between Au and Cu films prepared with the same protocol. Patch-dominated forces would therefore change by 20–200% under material swap.

7.2 Protocol

Step	Action
1	Measure $F(d)$ with Au surfaces at $d = 1, 10, 100$ μm
2	Swap to Cu (identical geometry, identical cleaning protocol)
3	Compare $\Delta F/F$
4	Apply decision criteria below

7.3 Decision Criteria

$\Delta F/F$ at $d = 10$ μm	Interpretation
< 5%	Consistent with Casimir prediction; patches subdominant
5–20%	Mixed regime; requires KPFM characterization for interpretation
> 20%	Patch-dominated; published Casimir extraction is provisional

To our knowledge, no published long-range ($d > 3$ μm) Casimir experiment has reported a material-swap comparison with quantitative $\Delta F/F$. We recommend it as a prerequisite for future claims in this regime. The test is independent of the $V^2\lambda^2$ degeneracy in §6 because it discriminates on the basis of a property the patch model predicts qualitatively (large material-dependence of variance) versus a property the Casimir prediction excludes quantitatively (sub-percent material-dependence).

8. Limitations and Scope

(i) We treat only electrostatic patches. Gravitational gradients (Paper II) and thermal-mechanical mode coupling (Paper III) are companion noise sources requiring independent treatment. (ii) Standard Lifshitz theory with Drude low-frequency extrapolation is assumed; the plasma-Drude controversy [12] modifies thermal contributions at the factor-of-two level, subdominant to patch-induced variations. (iii) The exponential correlation model captures short-range disorder; alternative spectra (Gaussian, power-law) preserve scaling laws and degeneracy structure. (iv) We do not claim impossibility — atomically-flat in-situ-cleaved single crystals with active KPFM nulling ($V_{\text{rms}} < 0.3 \text{ mV}$, $\lambda_c < 0.2 \text{ }\mu\text{m}$) reduce patch/Casimir to < 0.01 at $d = 10 \text{ }\mu\text{m}$. (v) Superconducting plates do not provide additional electrostatic screening for static patches beyond normal metals; this point is sometimes confused [15]. (vi) Surface roughness, contamination layers, and humidity effects introduce additional 10–50% systematics not modeled here.

9. Conclusions

We have presented a quantitatively verified noise budget for electrostatic patch contributions to long-range Casimir force measurements. The corrected Casimir pressures and patch/Casimir ratios are tabulated for parallel-plate geometry and discussed for sphere-plane geometry under PFA. We identify a likelihood degeneracy along $V_{\text{rms}}^2 \lambda_c^2$ that prevents unique parameter extraction from force data alone, and we propose a falsifiable material-swap test discriminating Casimir-dominated from patch-dominated regimes by an order-of-magnitude margin.

Re-examination of the most cited sub-micron and cryogenic Casimir measurements (Decca et al., Sushkov et al.) finds neither in a patch-dominated regime under standard parameter assumptions; this analysis therefore strengthens rather than contests those results. The danger zone for future long-range claims is identified as room-temperature sphere-plane geometry at $d \sim 3\text{--}10 \text{ }\mu\text{m}$ with standard-grade surfaces, where patches and Casimir become comparable and the $V^2 \lambda^2$ degeneracy becomes most severe. Spatially-resolved KPFM characterization and material-swap verification should be standard prerequisites for claims in this window.

Papers II (gravitational gradients and seismic noise) and III (thermal-mechanical mode coupling) of the Long-Range Casimir Series complete the noise-budget program for this regime.

Acknowledgments

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